

Amazon ECS Cluster Setup with EC2 Infrastructure

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon ECS cluster** was created from the **AWS Management Console** and the role of EC2 infrastructure was explained. The video showed that ECS clusters created from the console always include **Fargate** and **Fargate Spot** by default, and it then demonstrated how to add **EC2 infrastructure** during cluster creation. When EC2 is added, AWS automatically creates the required **Auto Scaling Group**, **launch template**, and **ECS-optimized EC2 instance** setup. The lab verified that the EC2 instance becomes part of the ECS cluster infrastructure and can be scaled by adjusting the Auto Scaling Group desired capacity. It also mentioned that if EC2 is not selected during creation, you can still add it later.

Understanding ECS Cluster Creation

An **Amazon ECS cluster** is the logical place where your containers run. When you create a cluster from the AWS Console, ECS gives you built-in serverless options and also lets you add EC2 capacity for more control. The newer UI shows three main infrastructure choices:

- **Fargate only**
- **Fargate and Managed Instances**
- **Fargate and Self-managed instances**

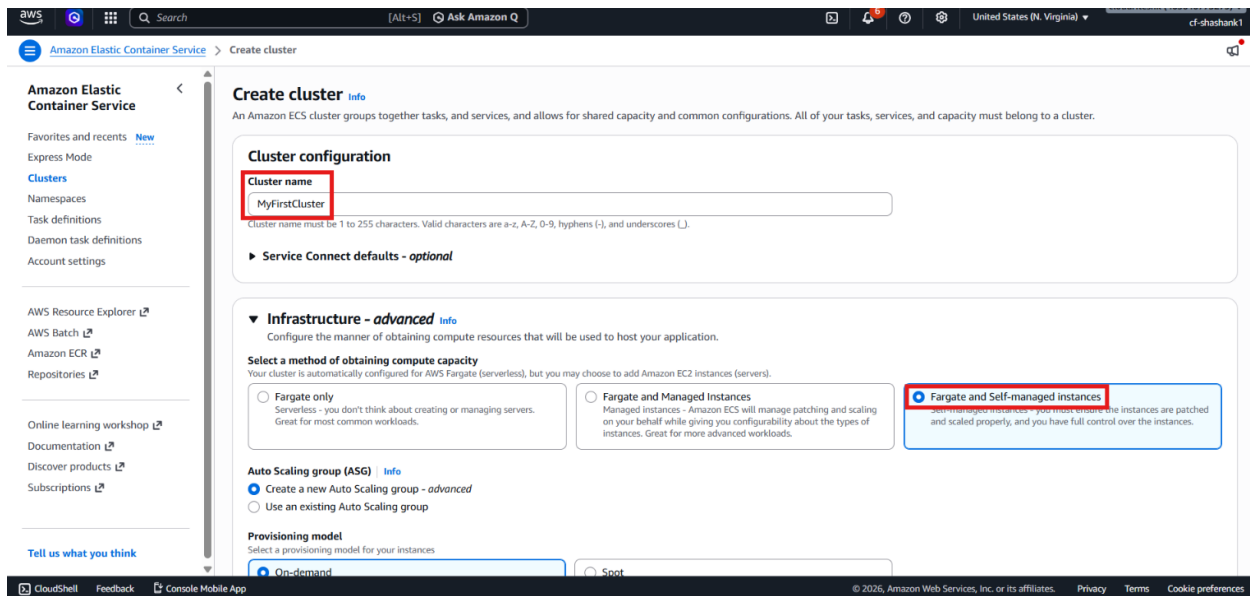
Fargate only is serverless, so AWS manages the compute layer. **Managed Instances** means AWS helps manage the EC2 layer for you. **Self-managed instances** means you control the EC2 instances yourself and attach them to the ECS cluster.

The lab focused on creating a cluster with EC2 added during cluster creation. ECS then created the supporting infrastructure automatically, including the Auto Scaling Group and the ECS EC2 instance registration.

Lab Steps

Step 1 – Open ECS and Create a Cluster

- Sign in to the AWS Management Console.
- Open **Amazon ECS**.
- Enter the cluster name: MyFirstCluster
- Review the infrastructure options shown in the current UI:
 - **Fargate only**
 - **Fargate and Managed Instances**
 - **Fargate and Self-managed instances**
- Choose **Fargate and Self-managed instances**



Step 2 – Configure the EC2 Infrastructure

- Choose to create a **new Auto Scaling Group**.
- Select the provisioning model:
 - **On-demand** for predictable usage
 - **Spot** for lower-cost spare capacity
- Choose an **ECS-optimized AMI** and for that i have selected **Amazon Linux 2 (kernel 5.10)**.
- Select an EC2 instance type, such as **t2.micro**.
- Create a new default role for now and attach the EC2 IAM role if you have already created one.
- Provide the desired instance count:
 - Minimum : 1
 - Maximum : 2
- Choose the key pair if already created otherwise create a new one.

Amazon Elastic Container Service > Create cluster

Provisioning model
Select a provisioning model for your instances.
☒ On-demand: On-demand instances, you pay for compute capacity by the hour, with no long-term commitments or upfront payments.
☐ Spot: Amazon EC2 Spot instances let you take advantage of unused EC2 capacity in the AWS cloud. [Learn more](#)

Container instance Amazon Machine Image (AMI)
Choose the Amazon ECS-optimized AMI for your instance.
Amazon Linux 2 (kernel 5.10)

EC2 instance type
Choose based on the workloads you plan to run on this cluster.
t3.micro
2 vCPU 1 GiB Memory

EC2 instance role
An instance role is used by Amazon EC2 instances to make AWS API requests. If you don't already have an instance IAM role created, we can create one for you.
Create default role [Create new instance profile](#)

Desired capacity
Specify the number of instances to launch in your cluster.
Minimum: 1 Maximum: 2

SSH Key pair
If you do not specify a key pair, you can't connect to the instances via SSH unless you choose an AMI that is configured to allow users another way to log in.
June-2026-us [Create a new key pair](#)

Root EBS volume size

- Select the default VPC and subnets.
- Select the security group.
- Enable public IP if needed.

A min of 30 GiB and a max of 16,384 GiB is allowed.

Network settings
The default VPC, subnets, and security groups are selected by default.

VPC
Select a VPC to use for your Amazon ECS resources.
vpc-099792461c7d0a616 [Create a new VPC](#)

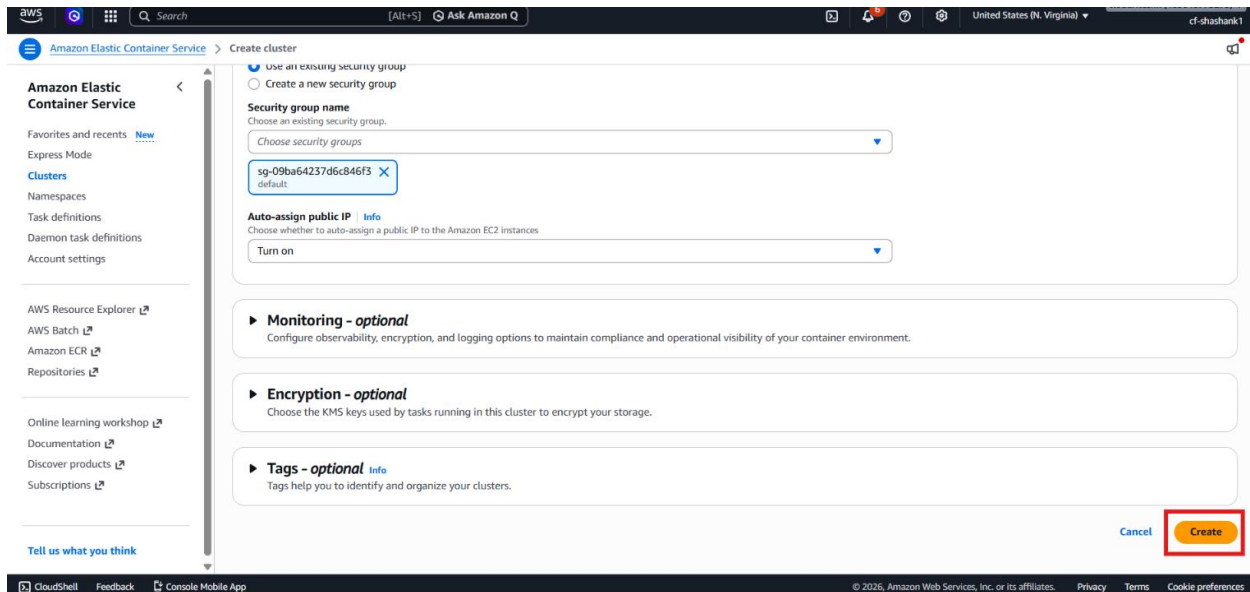
Subnets
Select the subnets where your instances are launched and your tasks run. We recommend that you use three subnets for production.
Choose subnets [Clear current selection](#)
subnet-0d10d67b324b6c99 subnet-0b5783ef03796c671

Security group [Info](#)
Choose an existing security group or create a new security group.
☒ Use an existing security group
☐ Create a new security group
Security group name: Choose an existing security group.
sg-09ba64237d6c846f3 [default](#)

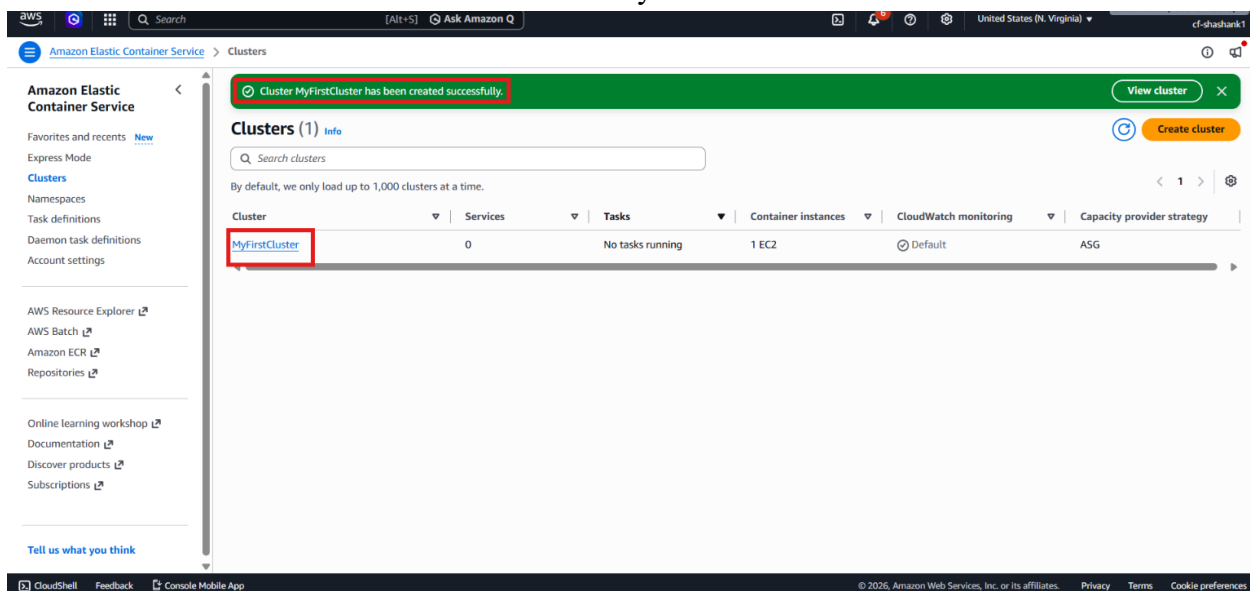
Auto-assign public IP [Info](#)
Choose whether to auto-assign a public IP to the Amazon EC2 instances.
Turn on

Step 3 – Create the Cluster

- Review the settings.
- Click **Create**.



- Wait for ECS to finish provisioning the infrastructure.
- Confirm that the cluster is created successfully.



Step 4 – Verify the ECS Infrastructure

- Open the cluster.
- Go to the **Infrastructure** tab.
- Confirm that:
 - Fargate is available
 - Fargate Spot is available
 - The EC2 infrastructure is present

Amazon Elastic Container Service

MyFirstCluster ASG

Cluster overview

Cluster ARN: [redacted] Status: Active CloudWatch monitoring: Default Registered container instances: 1

Services: 0 active | 0 draining Tasks: 0 running | 0 pending

Services | Tasks | Daemons | **Infrastructure** | Metrics | Scheduled tasks | Configuration | Event history | Tags

Capacity providers (3) Info

Filter capacity providers by property or value

Capacity provider	Scaling type	Provisioning model	Update status	Update status reason
☐ Fargate	Fargate	-	-	-
☐ Fargate Spot	Fargate Spot	-	-	-
☐ Infra-ECS-Cluster-MyFirstCluste...	EC2 Auto Scaling	-	-	-

Container instances (1) Info

Filter container instances by property or value

○ The Auto Scaling Group exists

EC2 > Auto Scaling groups

Auto Scaling groups (1) Info

Search your Auto Scaling groups

Name	Status - new	Launch template/configuration	Instances	Instance health - new	Desired capacity	Min
Infra-ECS-Cluster-MyFirstCluster-0198a6c3-ECSAutoScalingGroup-f8CIVPT5nzEX	At desired capacity	ECSLaunchTemplate_qCzuVKS5Y3QE Ver 1	1	1/1 Healthy	1	1

0 Auto Scaling groups selected

○ The EC2 instance is registered in the cluster

Step 5 – Scale the EC2 Capacity

- Open the Auto Scaling Group.
- Edit the desired capacity.

The screenshot shows the AWS Management Console for an Auto Scaling group named 'Infra-ECS-Cluster-MyFirstCluster-0198a6c3-ECSAutoScalingGroup-f8CIVPT5nzEX'. The 'Capacity overview' tab is active, displaying a progress bar for capacity and a table with the following details:

Desired capacity	Scaling limits	Desired capacity type	Status
1	1 - 2	Units (number of instances)	At desired capacity

Below this, the 'Launch template' section shows details for the 't3.micro' instance type, including AMI ID, security groups, and storage. An 'Edit' button is also present in the top right of this section.

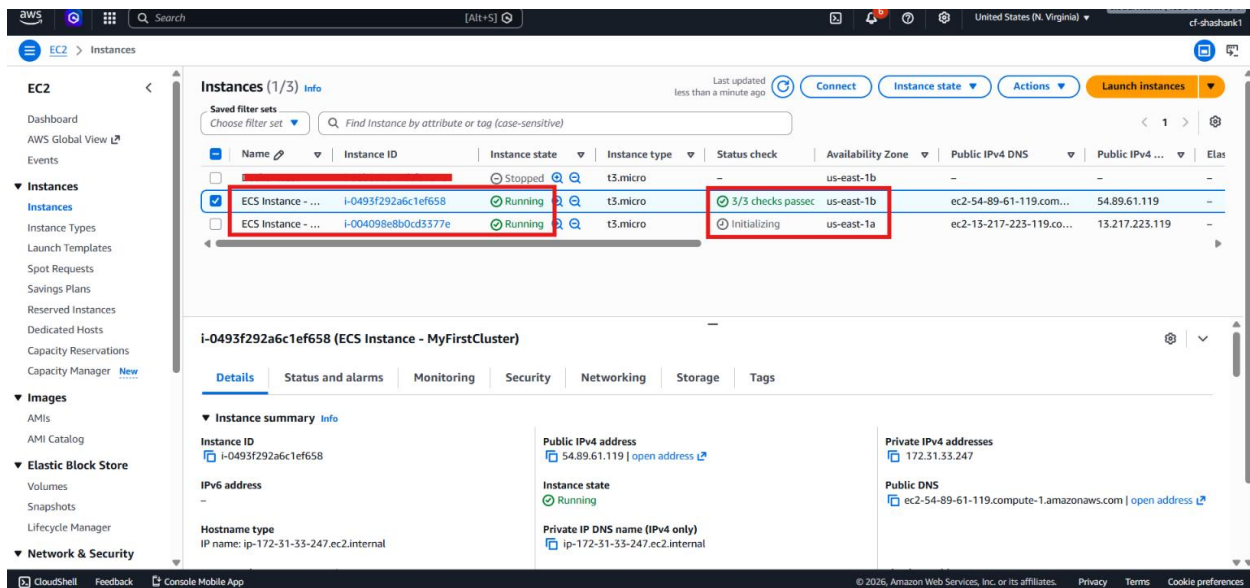
- Increase the number of instances if needed.
- Save the update.

The screenshot shows the same AWS Management Console page, but with the 'Update' dialog box open. The dialog box contains the following information:

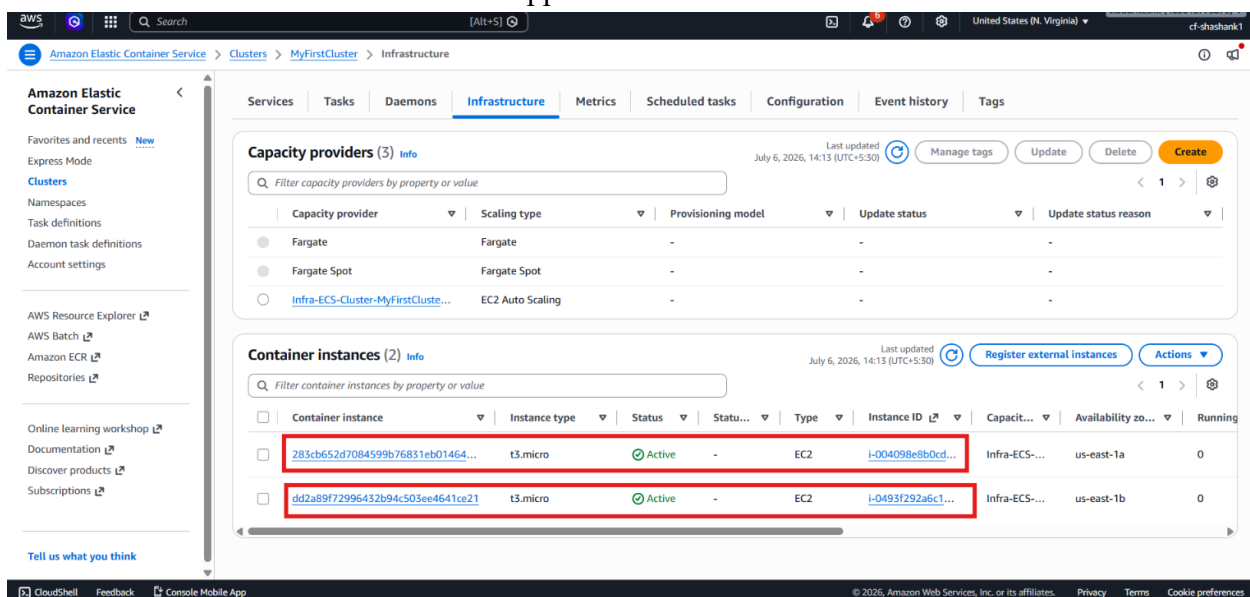
- Group size:** Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify minimum and maximum scaling limits.
- Desired capacity type:** Choose the unit of measurement for the desired capacity value. vCPUs and Memory (GiB) are only supported for mixed instances groups configured with a set of instance attributes. The dropdown is set to 'Units (number of instances)'.
- Desired capacity:** A text input field showing '2'.
- Scaling limits:** Set limits on how much your desired capacity can be increased or decreased. It includes fields for 'Min desired capacity' (set to 1) and 'Max desired capacity' (set to 2).

The 'Update' button is highlighted in orange at the bottom right of the dialog box.

- Wait for the new EC2 instance to launch.



- Refresh the ECS cluster infrastructure tab.
- Confirm that the new EC2 instance appears in the cluster.



Verification

- The ECS cluster was created successfully.
- EC2 infrastructure was added during cluster creation.
- ECS created the Auto Scaling Group automatically.
- The ECS-optimized EC2 instance registered with the cluster.
- The cluster infrastructure tab showed the EC2 instance.
- Increasing the Auto Scaling Group capacity added another instance successfully.